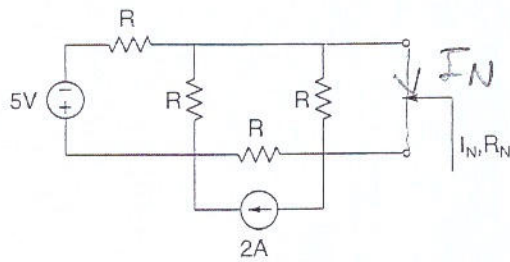
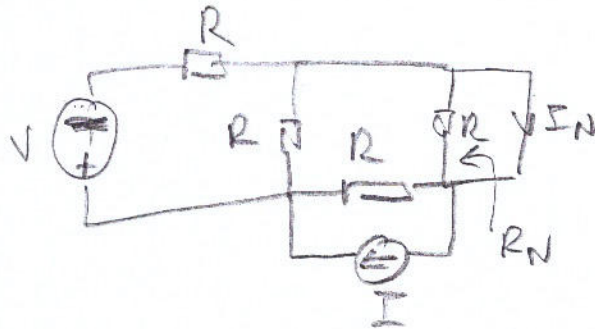


Question 1a [10 Points]:



For the circuit given above, find the Norton equivalent circuit. Draw the equivalent circuit.

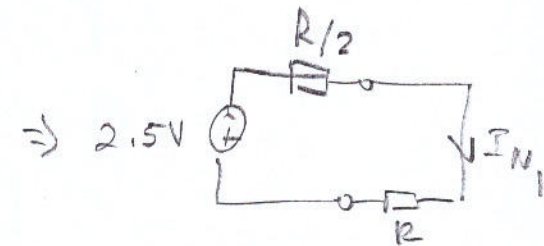
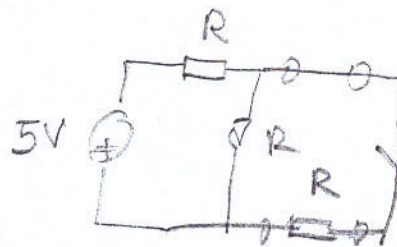


$$R_N = R \parallel (R + R \parallel R)$$

when $I=0$; $V=0$

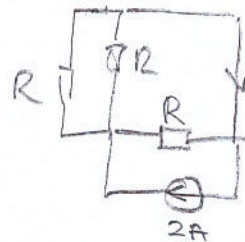
$$R_N = R \parallel (R + R/2) = \frac{3}{5} R$$

I_N : (i) $I=0$



$$I_{N1} = \frac{-2.5V}{R + \frac{R}{2}} = -\frac{5V}{3R}$$

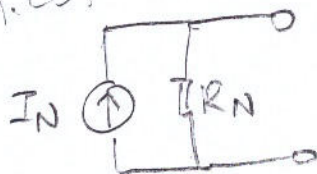
(ii) $V=0$



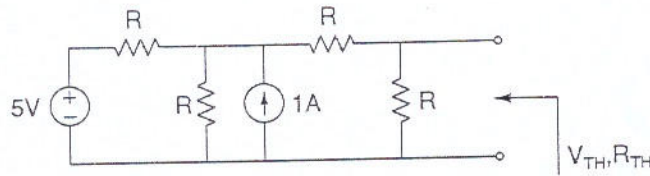
$$\frac{R}{R + \frac{R}{2}} = \frac{4}{3} A$$

$$I_N = I_{N1} + I_{N2} = -\frac{5}{3R} + \frac{4}{3}$$

eq. cat



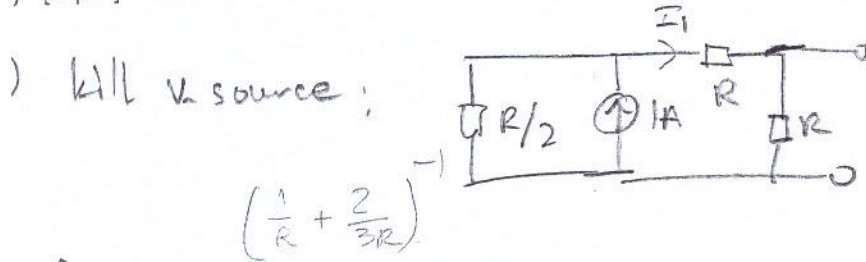
Question 1b [10 Points]:



For the circuit given above:

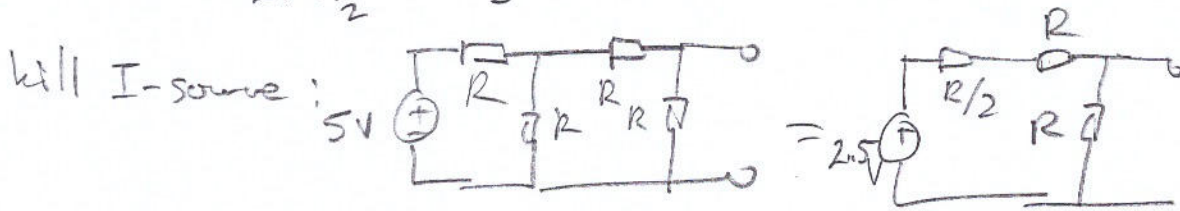
[5 pts] Find the Thevenin equivalent circuit using superposition. Draw the equivalent circuit.

[5 pts] Find the Thevenin equivalent circuit using node analysis. Draw the equivalent circuit.



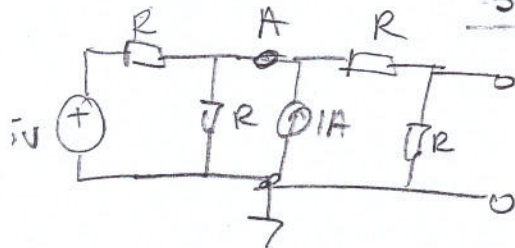
$$R_{TH} = R \parallel \left(R + \frac{R \parallel R}{2} \right) = \frac{3}{5} R$$

$$I_1 = 1A \cdot \frac{R/2}{2R + R/2} = \frac{(1A)}{5} \Rightarrow V_{oc1} = R I_1 = \frac{R}{5} V$$



$$V_{oc2} = (2.5V) \frac{R}{R + R + R/2} = \frac{5}{5} V = 1V$$

$$V_{TH} = V_{oc1} + V_{oc2} = \frac{R}{5} V + 1V$$



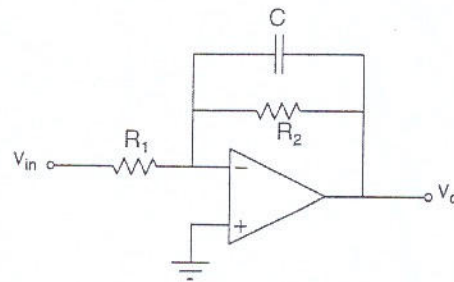
1 Node: $\frac{V_A - 5}{R} + \frac{V_A}{R} - 1 + \frac{V_A}{2R} = 0$

$$2V_A - 10 + 2V_A - 2R(1A) + V_A = 0$$

$$5V_A = 2R(1A) + 10 \Rightarrow V_A = \frac{2R}{5} + 2V$$

$$V_{TH} = \frac{1}{2} V_A = \frac{R}{5} + 1V$$

Question 2 [20 Points]:

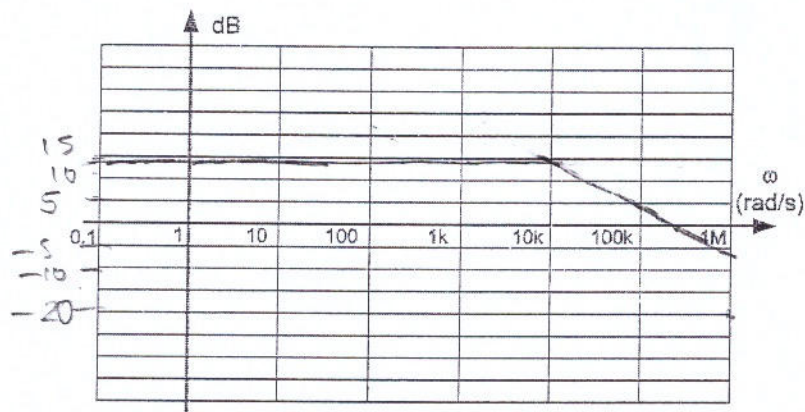


For the circuit given above, let $C = 1\mu\text{F}$. Assume that $v_{in}(t)$ is sinusoidal and the OPAMP is not saturated.

- (a) [15 pts] Find R_1 and R_2 such that the transfer function of the circuit is equal to:

$$H(\omega) = \frac{V_o(\omega)}{V_{in}(\omega)} = -\frac{5}{1 + j\omega 10^{-4}}$$

- (b) [5 pts] On the axis provided below, plot the low-frequency and high-frequency asymptotes of $|H(\omega)|$ in dB.



$$(a) H(\omega) = -\frac{\left(\frac{1}{R_2} + j\omega C\right)^{-1}}{R_1} = -\frac{R_2}{R_1} \frac{1}{1 + j\omega C R_2}$$

$$\therefore \frac{R_2}{R_1} = 5 \Rightarrow R_2 = 5R_1$$

$$\text{and } C R_2 = 10^{-4} \Rightarrow R_2 = 10^{-4} / C = 10^{-4} / 10^{-6} = \underline{100\Omega}$$

$$\therefore R_1 = 20\Omega$$

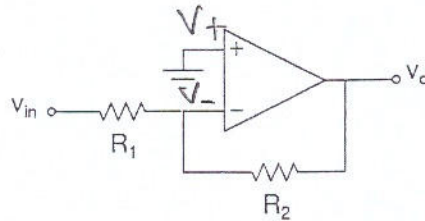
$$2) 20 \log |H(\omega)| = 20 \log 5 - 20 \log [1 + (\omega 10^{-4})^2]^{1/2}$$

$$\text{LF asymptote} \approx 20 \log 5 \approx 20(0.7) = 14 \text{ dB}$$

$$\text{HF asymptote} : -20 \log (\omega 10^{-4}) \quad \text{for } \omega 10^{-4} \gg 1$$

$$-3 \text{ dB} : \omega 10^{-4} = 1 \Rightarrow \omega = 10^4 \text{ rad/sec}$$

Question 3 [20 Points]:



For the circuit given above, assume that $v_{in}(t)$ is sinusoidal and the OPAMP is not saturated. In the questions below, A is the open-loop voltage gain of the OPAMP.

(a) [5 pts] If $A = \infty$, find $H(\omega) = \frac{V_o(\omega)}{V_{in}(\omega)}$.

(b) [5 pts] If $A = 10^5$, find $H(\omega) = \frac{V_o(\omega)}{V_{in}(\omega)}$.

(c) [5 pts] If $A = \frac{10^5}{1+j\frac{\omega}{50}}$, find $H(\omega) = \frac{V_o(\omega)}{V_{in}(\omega)}$.

(d) [5 pts] In (c), what is the frequency in Hz where $|H(\omega)|$ is 3-dB lower than its maximum value? Assume that $R_2 = 9R_1$.

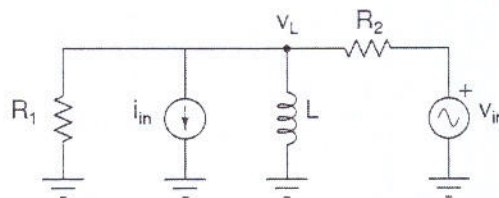
(a) $A = \infty$ $V_+ = V_- = 0$: $\frac{V_{in}}{R_1} + \frac{V_o}{R_2} = 0 \Rightarrow V_o = -\frac{R_2}{R_1} V_{in}$
 $H(\omega) = \frac{V_o(\omega)}{V_{in}(\omega)} = -\frac{R_2}{R_1}$

(b) $A = 10^5$: $V_- = V_{in} \frac{R_2}{R_1 + R_2} + \frac{V_o}{R_1} \frac{R_2}{R_1 + R_2}$
 $V_o = A(V_+ - V_-) = -A V_- = -A V_{in} \frac{R_2}{R_1 + R_2} - A V_o \frac{R_2}{R_1 + R_2}$
 $V_o \left(1 + \frac{A R_2}{R_1 + R_2}\right) = -\frac{A R_2}{R_1 + R_2} V_{in} \Rightarrow \frac{V_o}{V_{in}} = -\frac{A R_2}{(R_1 + R_2) + A R_1} = -\frac{R_2}{R_1} \frac{1}{\frac{R_1 + R_2}{A R_1} + 1}$
 $\frac{V_o(\omega)}{V_{in}(\omega)} = -\frac{R_2}{R_1} \frac{1}{1 + \frac{1}{A} \left(1 + \frac{R_2}{R_1}\right)}$ i.e. if $\frac{R_2}{R_1} \ll 10^5$, $\frac{V_o}{V_{in}} \approx -\frac{R_2}{R_1}$

(c) $A = \frac{10^5}{1+j\frac{\omega}{50}}$: $\frac{V_o(\omega)}{V_{in}(\omega)} = -\frac{R_2}{R_1} \frac{1}{1 + \frac{1}{10^5} \left(1 + \frac{R_2}{R_1}\right) (1+j\frac{\omega}{50})}$
 $\frac{V_o(\omega)}{V_{in}(\omega)} \approx -\frac{R_2}{R_1} \frac{1}{\left(1 + \frac{1}{10^5} \frac{R_2}{R_1}\right) + j\omega \frac{(1 + R_2/R_1)}{50 \times 10^5}}$

(d) $\omega_{-3dB} = \frac{50 \times 10^5}{1 + \frac{R_2}{R_1}} \left(1 + \frac{1}{10^5} \frac{R_2}{R_1}\right)$ $\frac{R_2}{R_1} = 9 \Rightarrow \omega_{-3dB} \approx \frac{50 \times 10^5}{10} = 5 \times 10^5 \text{ rad/s}$

Question 4 [20 Points]:



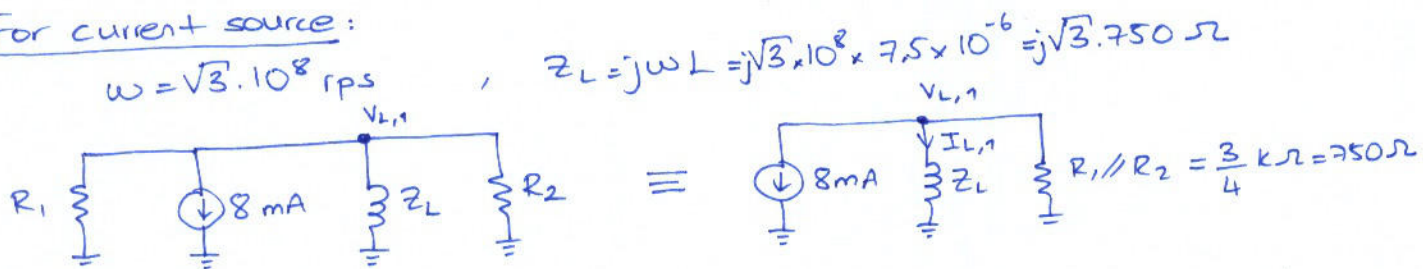
In the circuit above, $R_1 = 3 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, $L = 7.5 \text{ }\mu\text{H}$. The current and voltage sources are:

$$i_{in}(t) = 8 \cos(\sqrt{3} \cdot 10^8 t) \text{ mA} \quad \text{and} \quad v_{in}(t) = 4\sqrt{2} \cos\left(10^8 t + \frac{\pi}{3}\right) \text{ Volts.}$$

Find the voltage across the inductor, $v_L(t)$, using superposition.

Hint: Because the inputs are at different frequencies, superposition should be applied.

* For current source:



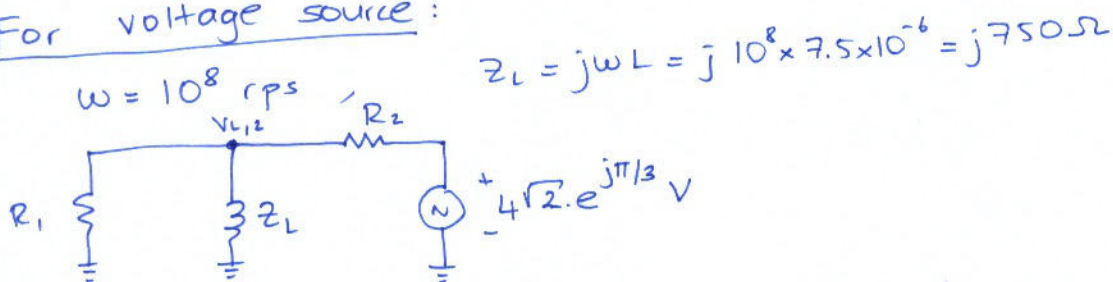
$$V_{L,1} = I_{L,1} \cdot Z_L = \left(- \frac{8 \times 10^{-3} (R_1 // R_2)}{Z_L + R_1 // R_2} \right) \cdot Z_L \quad (\text{from current divider})$$

$$= -8 \times 10^{-3} \cdot \frac{750}{j\sqrt{3}750 + 750} \cdot j\sqrt{3}750 = -j6 \cdot \frac{\sqrt{3}}{1 + j\sqrt{3}}$$

$$= \frac{6\sqrt{3} \cdot e^{-j\pi/2}}{2 \cdot e^{j\pi/3}} = 3\sqrt{3} e^{-j\pi/2 - j\pi/3} = 3\sqrt{3} e^{-j5\pi/6}$$

$$v_{L,1}(t) = 3\sqrt{3} \cos(\sqrt{3} \cdot 10^8 t - 5\pi/6) \text{ Volts}$$

* For voltage source:



$$V_{L,2} = \frac{Z_L // R_1}{Z_L // R_1 + R_2} \cdot V_{in} \quad (\text{voltage divider})$$

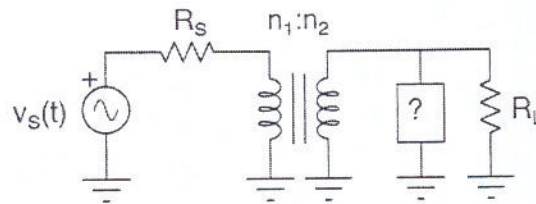
$$= \frac{j750 \times 3000}{3000 + j750} \cdot 4\sqrt{2} e^{j\pi/3} = \frac{j \cdot 3000}{4 + j} \cdot 4\sqrt{2} e^{j\pi/3}$$

$$= \frac{3j}{4 + 4j} \cdot 4\sqrt{2} e^{j\pi/3} = \frac{3 \cdot e^{j\pi/2} \cdot 4\sqrt{2} \cdot e^{j\pi/3}}{4\sqrt{2} \cdot e^{j\pi/4}} = 3 \cdot e^{j(\pi/2 + \pi/3 - \pi/4)} = 3 \cdot e^{j7\pi/12}$$

$$v_{L,2}(t) = 3 \cdot \cos(10^8 t + 7\pi/12) \text{ Volts}$$

* Using superposition: $v_L(t) = v_{L,1}(t) + v_{L,2}(t)$
 $= 3\sqrt{3} \cos(\sqrt{3} \cdot 10^8 t - 5\pi/6) + 3 \cos(10^8 t + 7\pi/12) \text{ V}$

Question 5 [20 points]:



We have a source with a source impedance of $R_s = 50 \, \Omega$ that supplies $v_s(t) = 5 \cos(2 \cdot 10^6 t) \, \text{V}$. The load impedance is $R_L = 10 \, \Omega$. We would like to use a transformer for impedance matching, to maximize the power delivered to the load. The inductance of the primary winding is measured to be $L_p = 10 \, \mu\text{H}$.

- (a) [6 pts] Find the turns ratio for the transformer that can achieve this matching.
- (b) [8 pts] For maximum power transfer to the load, what reactive component should we put in place of "?", and what should be its value?
- (c) [6 pts] If the inductance constant of the transformer core is given as $A_L = 25 \, \text{nH/turns}^2$, find n_1 and n_2 . If you find non-integer values, round them to closest integer values.

a) Ignoring "?" for now:

$$Z_{in} = \left(\frac{n_1}{n_2}\right)^2 Z_L \Rightarrow 50 = \left(\frac{n_1}{n_2}\right)^2 \cdot 10 \Rightarrow \boxed{\frac{n_1}{n_2} = \sqrt{5}} = 2.236$$

b) "?" should tune out inductance of the secondary at $\omega = 2 \cdot 10^6 \, \text{rps}$. $\Rightarrow$ use a capacitor

$$\left. \begin{aligned} L_s &= n_2^2 A_L \\ L_p &= n_1^2 A_L \end{aligned} \right\} \frac{L_s}{L_p} = \left(\frac{n_2}{n_1}\right)^2 = \frac{1}{5} \Rightarrow L_s = \frac{L_p}{5} = \boxed{2 \, \mu\text{H}}$$
$$C = \frac{1}{\omega^2 L} = \frac{1}{4 \cdot 10^{12} \cdot 2 \cdot 10^{-6}} = \frac{1}{8 \cdot 10^{-6}} = \boxed{0.125 \, \mu\text{F}} = \boxed{125 \, \text{nF}}$$

c) $L_p = n_1^2 A_L \Rightarrow 10 \times 10^{-6} = n_1^2 \cdot 25 \times 10^{-9}$

$$n_1^2 = \frac{10}{25} \cdot 10^3 = 400 \Rightarrow \boxed{n_1 = 20}$$

Then, $n_2 = \frac{n_1}{\sqrt{5}} = \frac{20}{\sqrt{5}} = 8.94 \approx \boxed{9}$